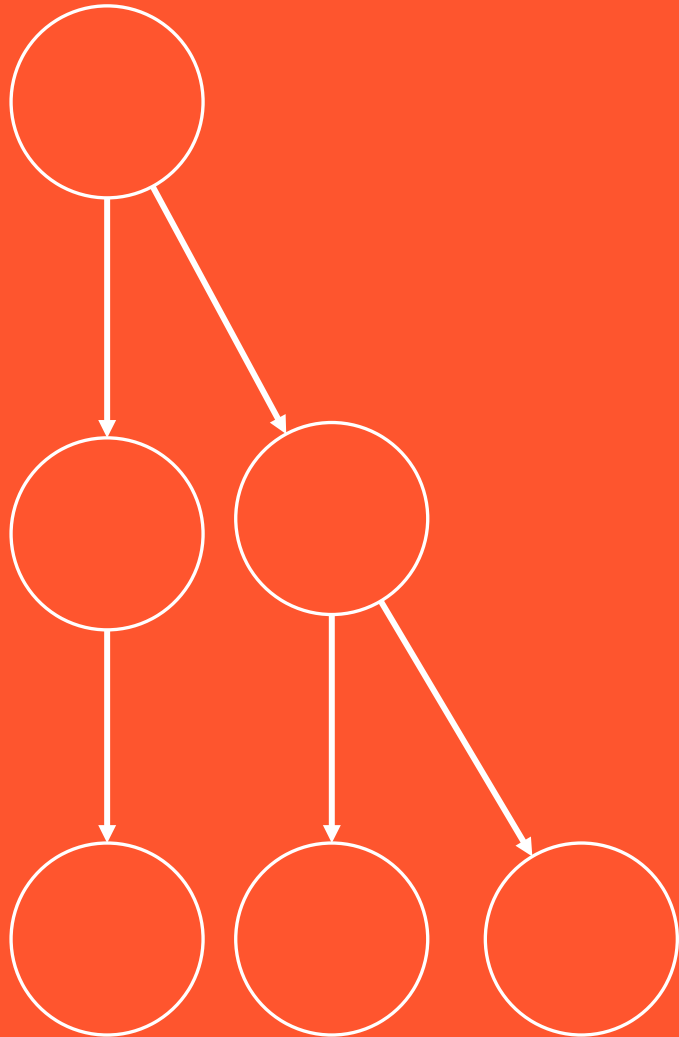




BCOG 100

Vertebrate Neural Architecture
Lecture 6.1 — Why Build Specialists



UNIVERSITY OF
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To Start: One Mechanism, Two Jobs

In your group, three minutes.

One mechanism that does everything is specialized for nothing.

Think of a tool, a machine, or a person asked to do two jobs, where getting better at one makes it worse at the other.

Should the jobs be split? What would splitting them cost?

A Living Jawless Fish

Not an ancestor: a modern animal whose lineage split from ours more than 500 million years ago. What both brains share was most likely in the common ancestor.

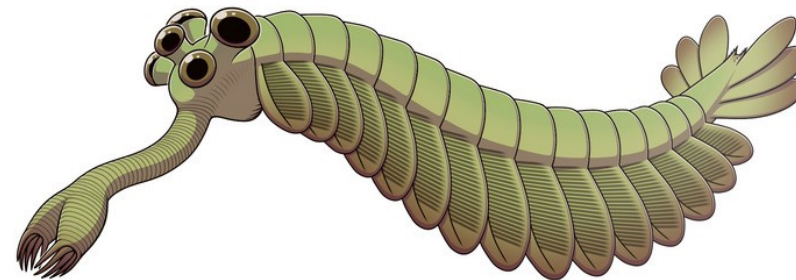


A Seafloor Where Nothing Hunted

Ediacaran (top): grazers on a microbial mat, food that lay still, no claws or shells. **Cambrian** (bottom): claws, and image-forming eyes to aim them.



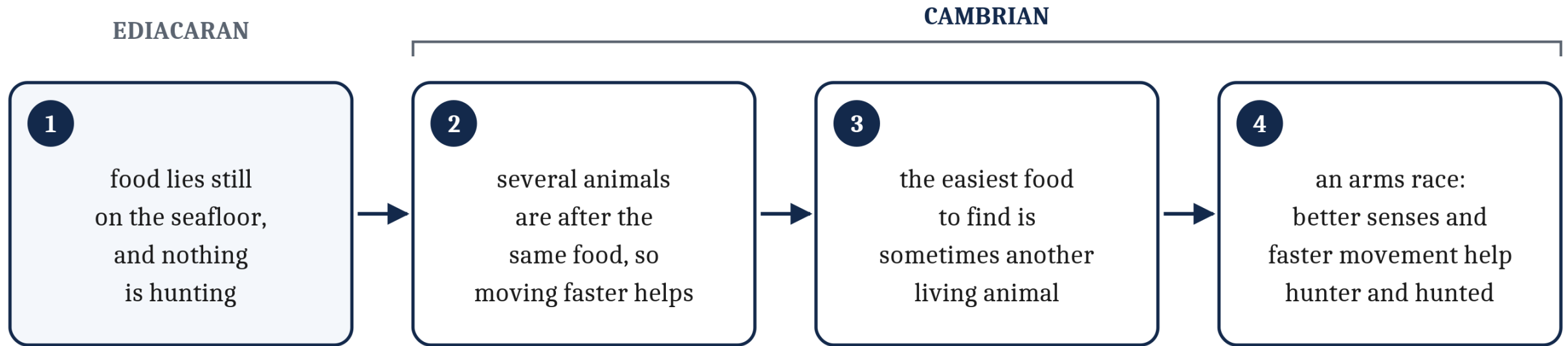
Jun (@ni075)



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Food That Stopped Lying Still

The change that matters for nervous systems is a sequence, and it ends in the arms race from Module 3. The **Cambrian explosion** is where the race started.



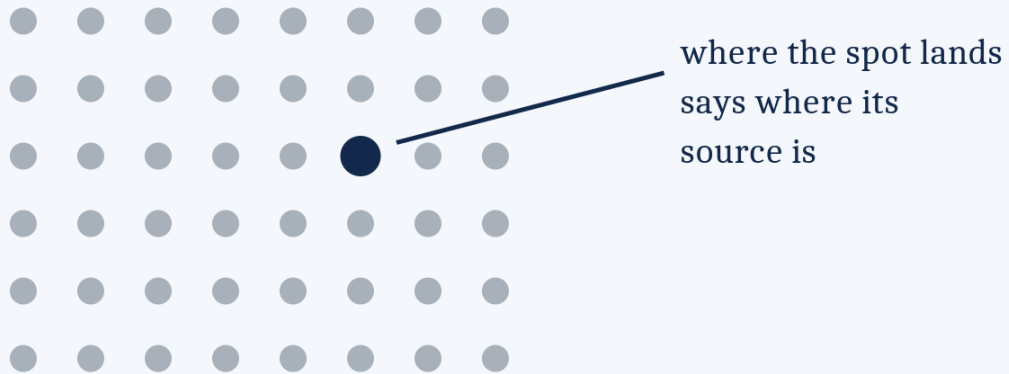
What is new is not the race. It is where the race started: with food that stopped lying still.

Why it happened when it did is still argued, and some researchers doubt it was an explosion at all.

What the Cambrian Asked of a Nervous System

Two new demands, and a third that is about the other two: they all arrived at once.

1 Senses became arrays



useless without processing: compare neighbors,
find edges, follow a shape as it moves

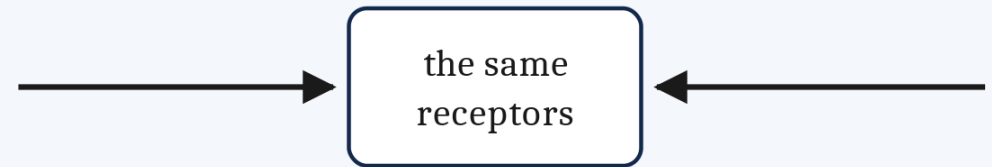
2 Tell self-caused from world-caused

self-caused:

the animal's own swimming

world-caused:

a predator's touch



a trickle for a slow worm;
a flood for a fast animal with jointed limbs

3 And they arrived together: find food, watch for predators, stay upright, keep the gills working.

Not whether a nervous system can meet one demand, but whether it can meet all of them at once.

Demands That Pull One Mechanism Apart

Improve a single mechanism for one side of each pair, and it gets worse at the other.



easy to switch means easy to interrupt; hard to interrupt means slow to switch



a fish cannot turn left and right at once, so the proposals have to meet and be weighed



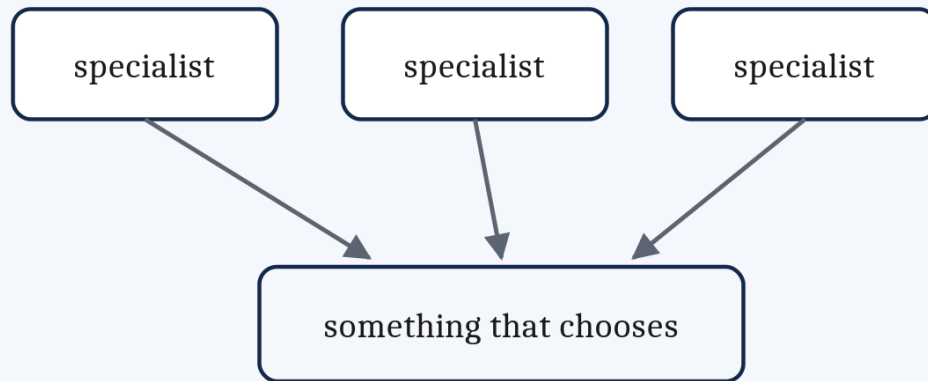
a mechanism that learns fast overwrites fast; one that keeps what it learned learns slowly

Improving one mechanism for one side of a pair makes it worse for the other.

Separate Structures Are One Answer Among Others

The demands are facts about the animal's world. Whether a brain meets them with separate parts is an **empirical question**.

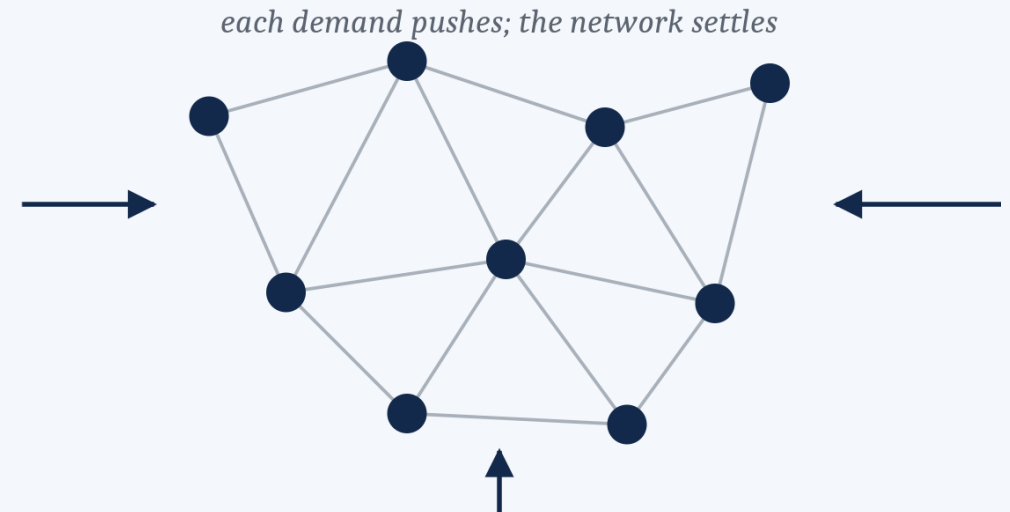
Separate structures



GAINS no interference; each part tuned to its job

COSTS wiring, coordination, and what one part learns is not shared

One network, soft constraints



GAINS shares everything it learns;
weighs everything at once

COSTS demands interfere; settling takes time

The demands are facts about the world. Which answer a nervous system took is an empirical question.

Real nervous systems sit between the two extremes.

What Specialists Cost

A brain's energy bill is set by how many neurons it has, not by how much it weighs. Wiring and coordination cost space and time as well.

Energy

**billed per neuron,
not per gram**

measured in mammals, mice to humans;
extended to fish by inference

Space

**every connection
takes room**

the longer the wire,
the more room it takes

Time

**long wires arrive late;
conflicts take time**

something has to settle
which output wins

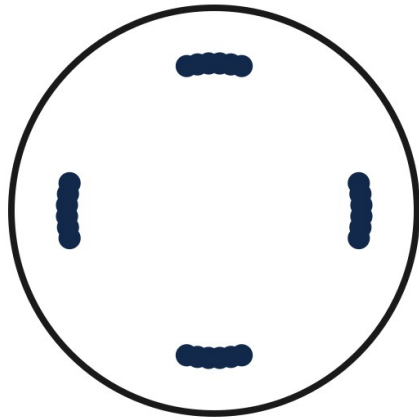
Energy can buy more neurons. It cannot buy back space or time.

A Body That Moves as One

A fish acts with its whole body, so its specialists sit together in one organ in the head: short wires, and one place for proposals to meet.

Gathered at the front, then divided into parts

a ring with ganglia



24 neurons



gathered at the front



24 neurons



divided into parts



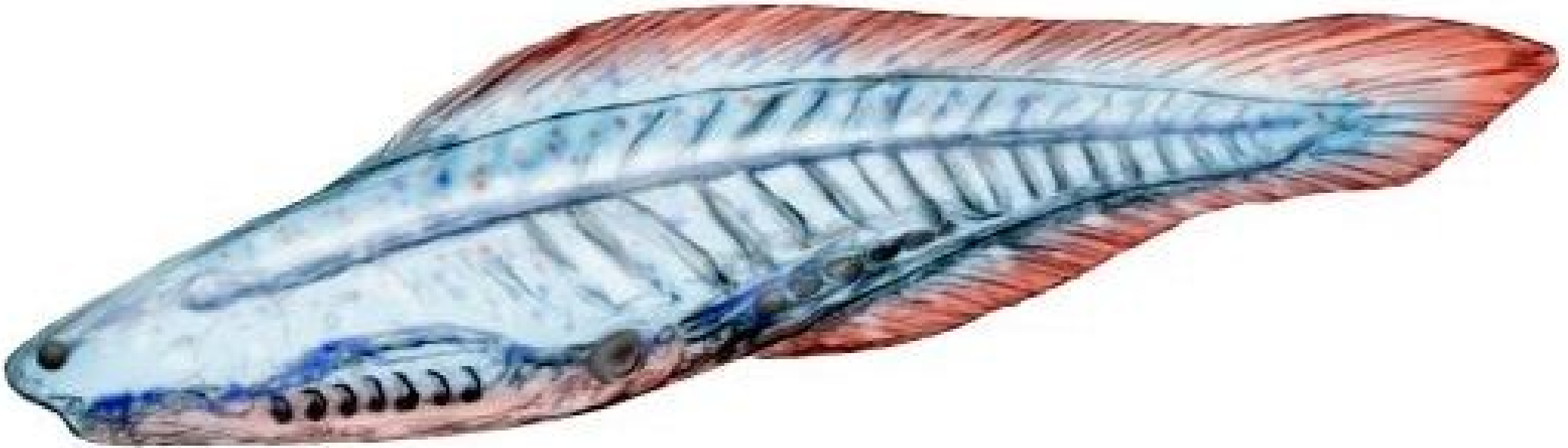
four parts,
each doing
its own job

the same neurons,
sorted into regions

gathering gets the neurons to one place; dividing gives each job its own

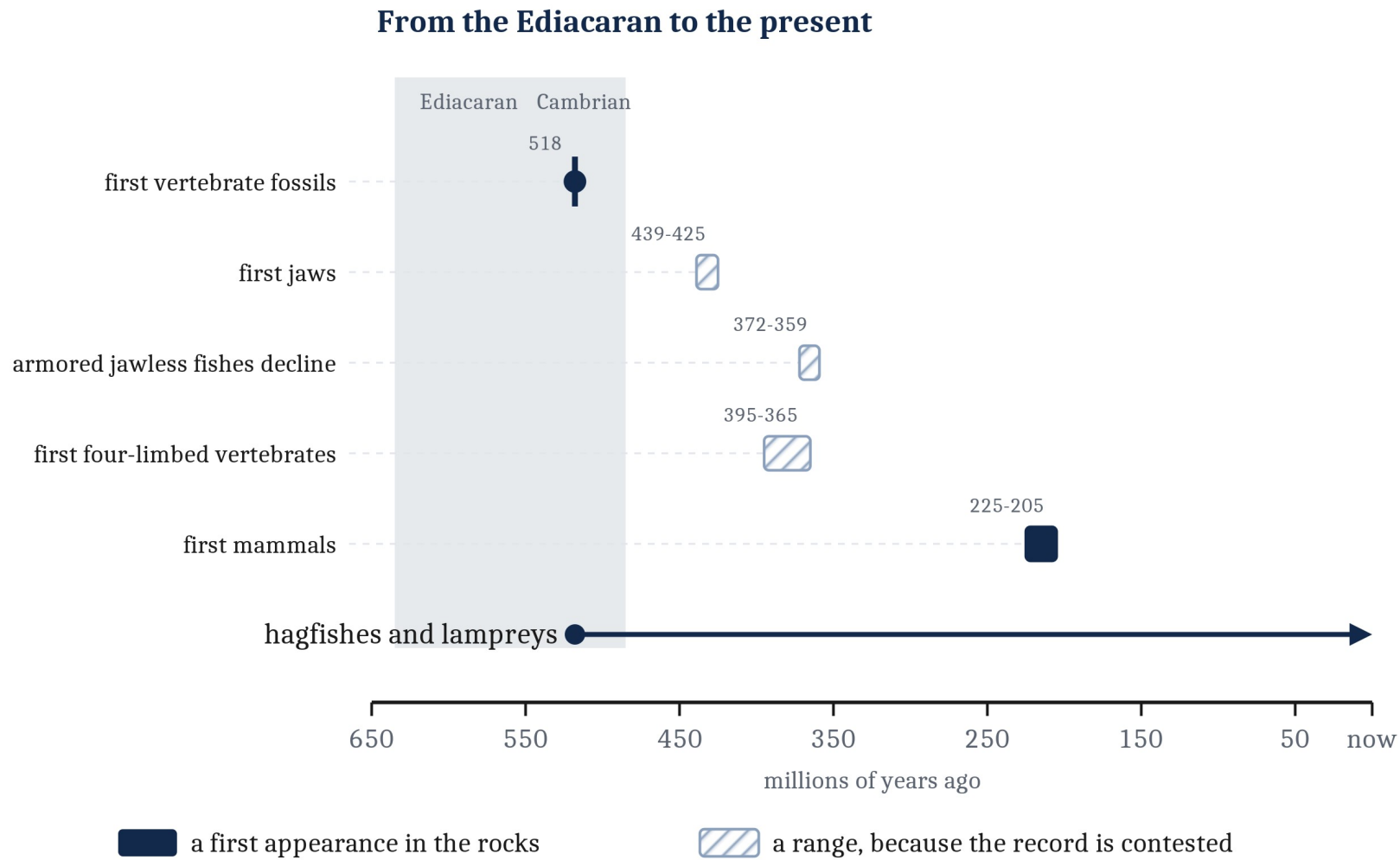
Slivers in the Sea

The first **vertebrates** were a few centimeters long, toothless, and most likely prey. What was new was a head, built largely by the **neural crest**.



Slivers in the Sea

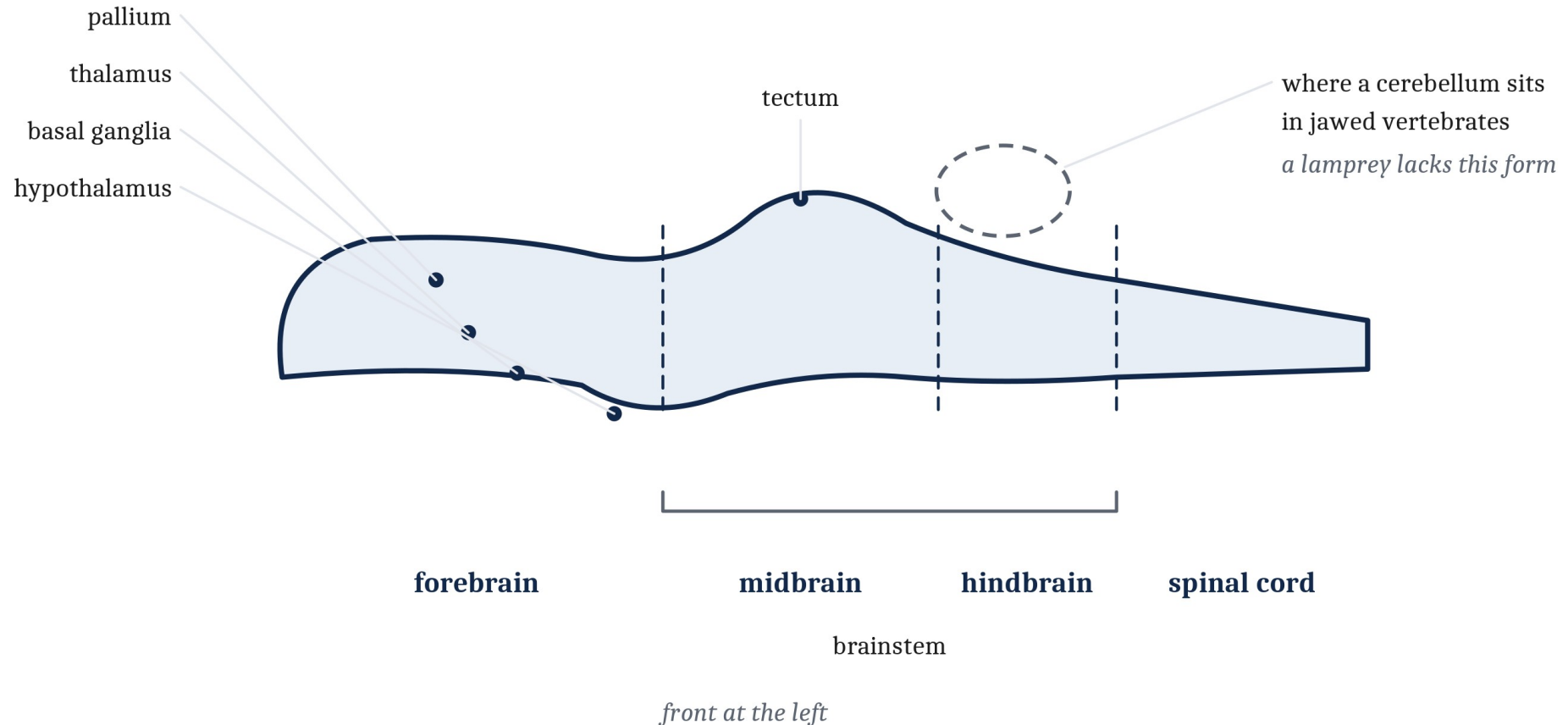
The first **vertebrates** were a few centimeters long, toothless, and most likely prey. What was new was a head, built largely by the **neural crest**.



The Parts of a Vertebrate Brain, in a Lamprey

Nearly every part the module is about is already here. The conspicuous absence is the **cerebellum**, in the form jawed vertebrates have it, which arose after the lamprey's lineage split off.

The brain of a lamprey, from the side



New Structures, Old Parts

Vertebrates did not invent any of these capacities. They added structures given over to them, built from material that was already there.

OLD

The capacities

approaching and avoiding, arousal, states that outlast their causes,
learning from reward, choosing one action over another

all in Modules 4 and 5, in animals far simpler than a lamprey

NEW

The structures

parts given over largely to one kind of job, assembled near
the origin of the vertebrates, and kept in every lineage since

OLD

The material

genes that mark head-to-tail position, the kinds of cells neurons are,
and the ways cells divide, move and connect

the neural crest: old sets of genes, and one new set that links them

New structures, built from old parts.

Specialists in a Machine

In your group, four minutes.

Some large AI language models are one network, with every part used for every word. Others are split into many "expert" sub-networks, and a router sends each word to just a few of them.

Using the ledger from this lecture: what does each design gain, and what does each pay?

What in a fish plays the part of the router, and how is it different?

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